LAB 12

Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	Regime filter
TSC reference	A4, K4, K6
Duration	15 min
Folder	labs/Lab12-market-regimes/
Files	Lab12-market-regimes/regime_stage.py, pipeline/regime.py, Lab12-market-regimes/solution/regime.py
Mode	Runs ONLINE with your Alpaca paper keys in labs/.env, or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

The risk team suspects the desk's signals behave differently in calm and stressed markets. It asks for an unsupervised regime model that can switch signals off in stress - fitted walk-forward so that the gate never uses information from the future.

Objective

Build pipeline/regime.py: PCA on the universe's returns, K-means regimes on backward-looking SPY features, walk-forward regime labels, and a gate that forces signals flat in the stress regime, evaluated after costs.

Before You Start

- Open a terminal in the labs folder and activate your virtual environment.
- Have your AI coding assistant open (for example Claude Code) in the labs folder if you want to rebuild the code from the prompts; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

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Step-by-Step Instructions

Step 1: Build PCA and the regime features (K4)

Ask your assistant for the first part of pipeline/regime.py.

AI prompt - paste into your AI coding assistant
Create labs/pipeline/regime.py. regime_features(closes) builds backward-looking SPY features: vol_21 (annualised 21-day volatility), ret_21, ret_63, dist_sma200 (close / ta_sma(200) - 1) and corr_spy_tlt (63-day rolling correlation of SPY and TLT returns), dropping NaN rows. fit_regimes(train, k) fits StandardScaler + KMeans(n_clusters=k, n_init=20, random_state=42) and names clusters by average vol_21: 'Low vol', 'Moderate vol', 'High vol (stress)' for k=3, otherwise 'Regime i (vol rank)' with the highest named 'High vol (stress)'. In run() also fit sklearn PCA on standardised daily returns of config universe.trade (pipeline.ingest.load_closes) and print variance explained by PC1-PC4 and the loadings. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: fit_regimes returns a scaler, a fitted KMeans and a dict of names ordered from low to high volatility.

Step 2: Add walk-forward labels and the gate (K6, A4)

Ask your assistant for the walk-forward loop, regime_gate(), run() and main().

AI prompt - paste into your AI coding assistant
In labs/pipeline/regime.py add walk_forward_regimes(feat, k, first_year=2020): for each calendar year fit_regimes on data up to 31 December of the previous year only and label that year's days; regime_gate(target, regimes, blocked=('High vol (stress)',)) forces the target to 0 on blocked days (the regime is known at the close, so the gate acts on the next bar like any signal). run(k=3) for `python -m pipeline regime --clusters`: print [1] PCA, [2] full-sample cluster profiles (labelled hindsight, description only) with next-day SPY Sharpe and trend-filter Sharpe by regime, [3] walk-forward regime counts and the latest regime and gate state, [4] gated versus ungated CAGR, Sharpe, max drawdown, exposure and trades from 2020 for the configured strategy, trend_filter and dip_buy (backtest with config costs). Save _outputs/Lab12_regime_labels.csv and chart Lab12_regimes.png (PCA variance, SPY coloured by walk-forward regime, gated vs ungated equity). Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline regime --help shows --clusters.

Step 3: Run the regime stage and read the PCA (A4, K4)

Run: python -m pipeline regime --clusters 3 (same as: python3 Lab12-market-regimes/regime_stage.py).

Check: PCA on 9 assets x 2,010 days: PC1 56.8%, PC2 15.5%, PC3 10.3% (cumulative 82.5%). PC1 loads 0.33-0.43 on the equities and -0.09 on TLT: one broad equity factor; TLT and GLD are the diversifiers.

Step 4: Read the regime profiles (K4, K6)

Read section [2]. In which regime does the trend filter work, and where does it fail?

Check: Low vol 654 days (vol 12.7%), Moderate vol 786 days (vol 13.7%, stock-bond correlation -0.33), High vol (stress) 372 days (vol 30.4%). Trend-filter Sharpe 1.23 / 1.04 / -0.45 by regime.

Step 5: Check the walk-forward labels (K6)

Read section [3]. Why does the walk-forward model label far fewer stress days than the full-sample fit?

Check: 2020-2025: Low vol 1,061, Moderate vol 383, High vol (stress) 64 days; latest regime (2025-12-31) Low vol -> gate OPEN.

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Step 6: Evaluate the gate after costs (A4)

Read section [4] and _outputs/Lab12_regimes.png. Should the gate go into the live pipeline?

AI prompt - paste into your AI coding assistant
Using the gated versus ungated table from python -m pipeline regime, write a short recommendation to the risk team on whether to switch the regime gate on for ema_cross, trend_filter and dip_buy. Quote Sharpe and max drawdown, mention that the gate is itself a model fitted on few stress days, and say which test from Lab 13 should decide.

Check: ema_cross 0.91 -> 0.89 and trend_filter 0.89 -> 0.80 (gate hurts); dip_buy 0.63 -> 0.72 with max drawdown -17.98% -> -12.37% (gate helps) - so the gate is signal-specific.

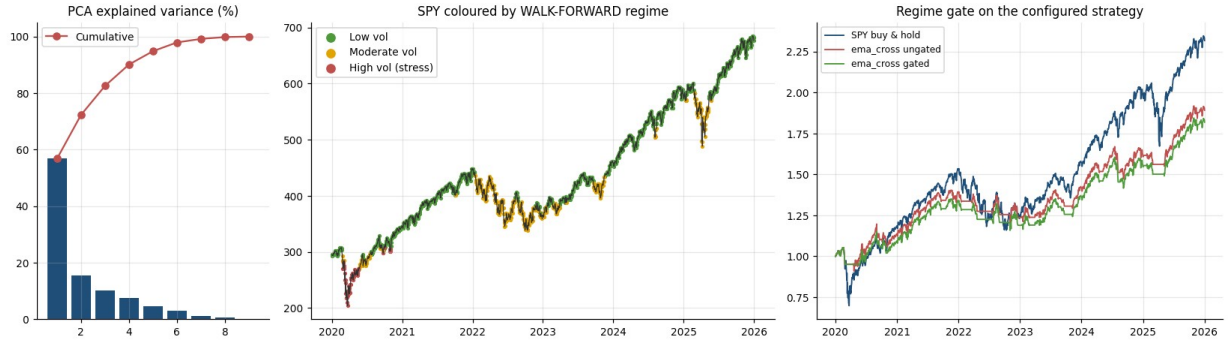
Step 7: Try four clusters (A4)

Run: python -m pipeline regime --clusters 4 and compare the profiles.

Check: Four regimes; the highest-volatility cluster has only 23 days (vol 80.8%, the March 2020 crash) - too few to trust a gate built on it.

Expected Output

python -m pipeline regime --clusters 3: PC1 56.8%, PC2 15.5%, PC3 10.3% (82.5% cumulative); regimes Low vol (654 days, vol 12.7%), Moderate vol (786, 13.7%), High vol (stress) (372, 30.4%); trend-filter Sharpe 1.23 / 1.04 / -0.45 by regime; walk-forward 2020-2025: 1,061 / 383 / 64 days, latest Low vol (gate open); gate effect on Sharpe: ema_cross 0.91 -> 0.89, trend_filter 0.89 -> 0.80, dip_buy 0.63 -> 0.72; files _outputs/Lab12_regime_labels.csv and chart _outputs/Lab12_regimes.png.



Lab 12 output (Lab12_regimes.png), offline mode on the bundled data.

Debrief

Unsupervised learning (K4) finds structure without labels: PCA compresses correlated assets into a few drivers and K-means groups days into regimes. Fitting the regimes walk-forward is part of validating the model (K6), and testing the gate signal by signal after costs (A4) shows that a filter can help one strategy and hurt another.

Troubleshooting

Symptom	What to do
Cluster names swap between runs	K-means numbers clusters arbitrarily; fit_regimes names them by volatility so labels stay meaningful.
ConvergenceWarning	Increase n_init or max_iter in KMeans; results are usually unaffected.
Walk-forward starts in 2020	The first fit needs at least 200 days of features after the 200-day warm-up.

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